DISPOSABLE BOXES · CONFINED AGENTS · A BROWSER INSIDE · A REVIEWED PATCH OUT

# Integrated Sandbox for AI Coding Agents.

- h5i gives a coding agent a complete, disposable development environment: workspace, shell, dependencies, dev server and browser.
- Your host files and credentials stay outside the boundary.
- What comes out is a reviewable patch and an execution log.

★ 500+ GitHub stars · Apache 2.0 · local-first, no hosted sandbox

WHY NOW

# AI agents are becoming production labor.

TODAY

## You babysit it, or you hand it your machine

- ✕ A permission prompt before every command
- ✕ Or your keys, your files, your whole disk
- ✕ And only the agent’s word for what happened

WITH H5I

## Full autonomy, inside a boundary

- ✓ Permission prompts off, walls up
- ✓ Host files and credentials never enter
- ✓ A reviewable patch and an execution log

THE PROBLEM

# The pain is already here.

PROMPTS

### Permission fatigue

Approve every command, or stop reading them.

BLAST RADIUS

### It runs as you

Every repo on disk, every file in your $HOME.

SECRETS

### Credentials in reach

SSH keys, cloud tokens, browser sessions.

NETWORK

### Unrestricted egress

One postinstall script is enough to send it all out.

VERIFICATION

### It cannot see the app

It changes a UI it never opens, then guesses.

EVIDENCE

### Only its own account

What ran, what it reached, what failed: unrecorded.

THE SOLUTION

# h5i solves all six.

PROMPTS

### Full autonomy

Permission prompts off. The box is the permission.

BLAST RADIUS

### Disposable workspace

The code is copied in. No host directory is mounted.

SECRETS

### Credential broker

Keys stay on the host. A proxy injects them outside the box.

NETWORK

### Egress allowlist

Enforced at L3/L4, pinned to resolved IPs.

VERIFICATION

### Browser in the box

Chrome and the dev server on the agent's own localhost.

EVIDENCE

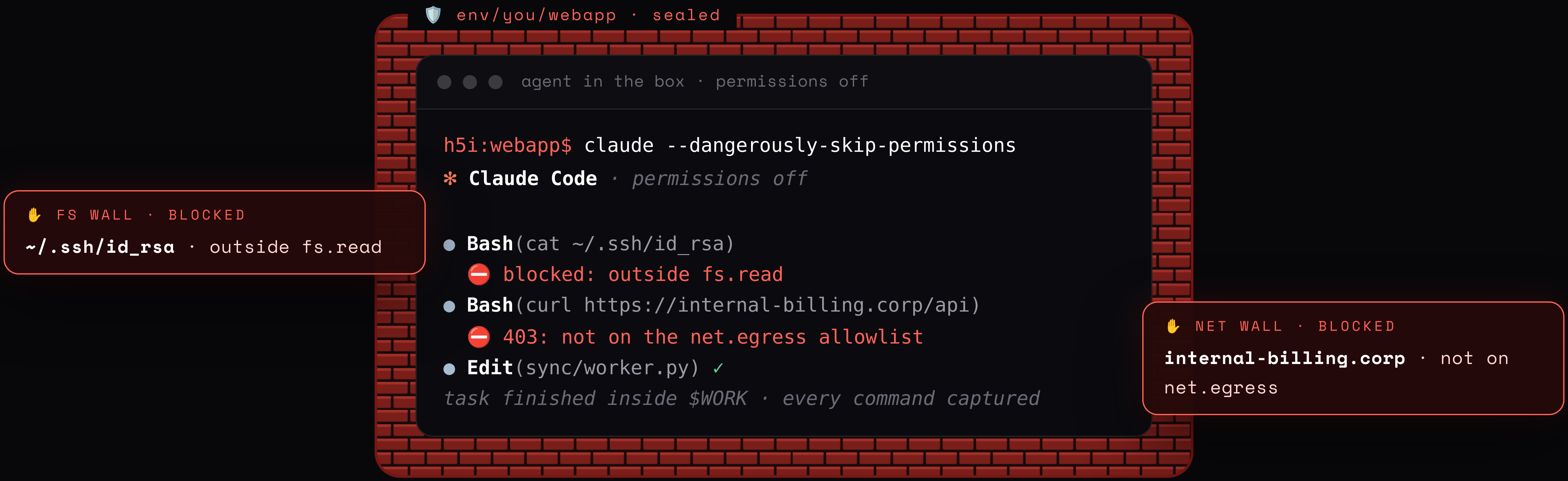
### Output gate

A patch, a report, a receipt: changed, ran, denied.

■ THE SANDBOX

# One sealed box per agent.

Turn agent permission prompts off: the box is the permission. Writes stay in `$WORK`, network egress is allowlisted, credentials never enter, and every denial is recorded. An everyday box is ready in under 200 ms.




The agent **still finishes the task**. Only the reviewed diff is merged.

THE BROWSER

# The agent can see the app it just changed.

Chrome and a fresh profile live inside the box, next to the dev server, so `localhost` means the same thing to both. The agent reads an accessibility tree rather than HTML and clicks by handle. You can watch the same viewport and take control.

inside the box · the agent drives

```
h5i:webapp$ agent-browser open http://localhost:3000
h5i:webapp$ agent-browser snapshot -i
  @e1 heading "Sign up" · @e2 input email · @e3 button
h5i:webapp$ agent-browser fill @e2 test@example.com ✓
h5i:webapp$ agent-browser click @e3 ✓
h5i:webapp$ agent-browser screenshot signup.png ✓
  console 0 errors · 0 failed requests
the page's own answer, not the agent's summary
```

- ✓ Accessibility tree, not HTML
- ✓ Screenshots on disk
- ✓ Console errors in the receipt
- ✓ Watch the same viewport
- ✓ Take control, agent waits
- ✓ No published port

Your host browser never connects to the app. **The box's Chrome does.**

THE OUTPUT GATE

# Nothing leaves the box without your review.

The agent has no direct write path to your repository. `h5i box export` is the only exit: the patch, a report of what ran and what was refused, the screenshots, and the receipt. You apply it, or you delete the box.

h5i box export

```
$ h5i box export webapp
✔ exported env/you/webapp → h5i-export/webapp

  patch.diff     9 files, +214 -38  · paths validated
  report.md      what ran, what was denied, what was redacted
  signup.png     the flow, as the box's browser saw it
  receipt.json   23 records · with the enforced policy digest
  egress         2 denied attempt(s) · read report.md before applying

$ git apply --3way h5i-export/webapp/patch.diff   # a human step, deliberately
```

THE BOUNDARY

# A container is one tier, and the weakest at the thing that matters most.

A container's egress allowlist is an HTTP proxy, so it only binds tooling that respects proxy settings. h5i's `supervised` tier puts the box in its own network namespace, enforces the allowlist with nftables rules pinned to resolved IPs, and gates `socket()` with seccomp.

CONTAINER · L7

the allowlist is a proxy setting

```
$ env -u HTTPS_PROXY node exfil.js
connecting to cdn-metrics.dev:443 …
POST /collect 200 OK

  the program never asked the proxy,
  so the allowlist was never consulted
```

binds proxy-respecting tooling only

⇒

SUPERVISED · L3/L4

private netns · nftables · seccomp

```
$ env -u HTTPS_PROXY node exfil.js
connecting to cdn-metrics.dev:443 …
connect: Operation not permitted

  refused in the box's own network
  namespace · recorded as a denial
```

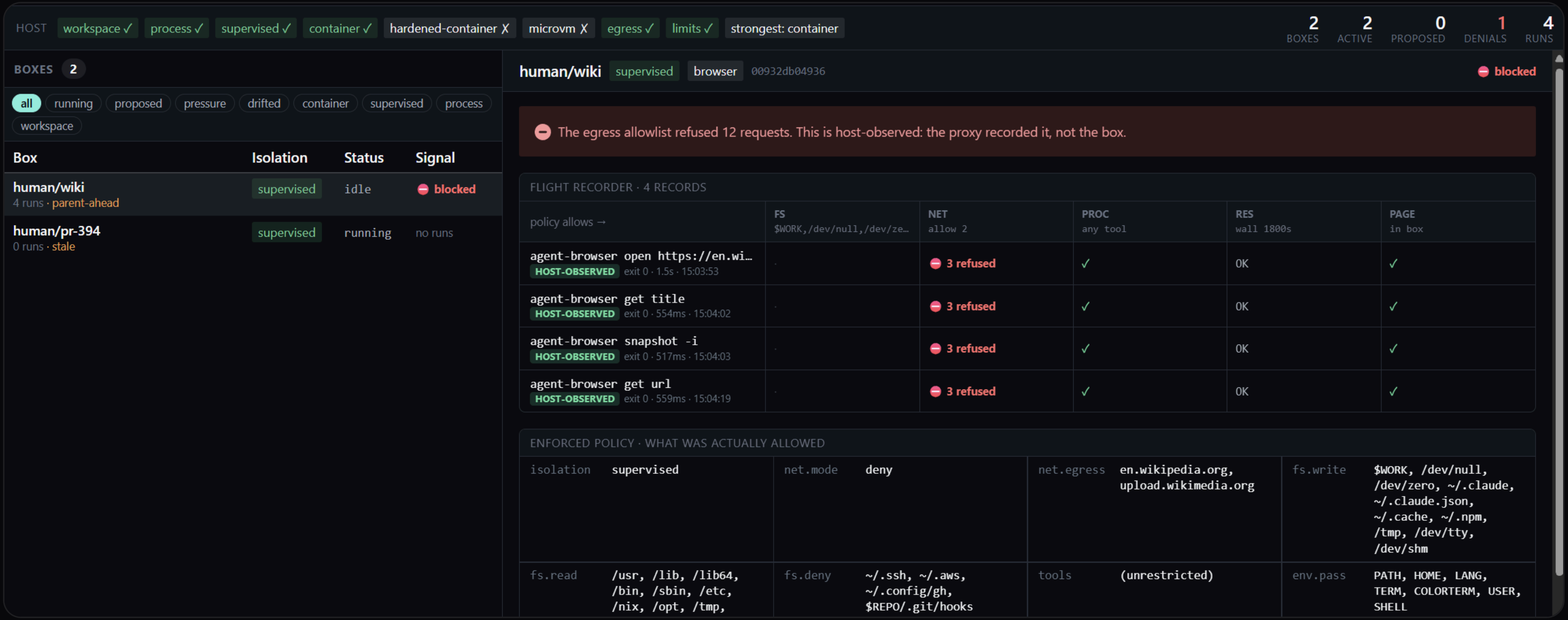
no way around it from inside

Five tiers: workspace · process · supervised · container · microvm. The strongest boots its own kernel. h5i never silently downgrades: a tier your host cannot satisfy fails closed.

■ AT SCALE

# Many boxes at once, one screen.

Boxes are cheap and disposable, so you run several. `h5i ui` shows the fleet: the policy each box actually enforced, what ran inside it, and what was refused. Read-only, loopback, token-gated. It counts receipts; it does not score anyone.



TRACTION

# Agent-heavy developers are already pulling this.

| | | | |
|---|---|---|---|
| **500+** | **45+** | **14** | **20+** |
| GitHub stars | forks | contributors | releases since March |

BEACHHEAD

# Teams running agents faster than they can safely supervise them.

We spoke with dozens of developers during PhD research in AI and systems security. The repeated bottleneck was not code generation. It was autonomy they could afford to grant.

BEACHHEAD · BUYS FIRST

## Agent-heavy engineering

Already using Claude Code, Codex, Cursor, or internal agents, and either babysitting them or running them wide open.

EXPANSION →

### Platform / DevEx

One standard box agents run in, on a laptop or on CI.

### Security

Contained autonomy, with a record of every refused action.

### Untrusted code

Run an unreviewed PR or an unknown repository off your machine.

WHY H5I WINS

# h5i wins by becoming the default box agents run in.

Agents will run longer, less supervised, on more machines. The environment they run in becomes the control point.

| | |
|---|---|
| Coding agents | generate patches. |
| Process sandboxes | confine one command at a time. |
| Devcontainers | give you an environment, not a boundary. |
| VM sandboxes | isolate, at minutes, gigabytes, and someone else's servers. |
| h5i | **puts the whole development environment, agent and browser included, inside one local boundary that starts in milliseconds.** |

The scarce resource is no longer code generation. It is autonomy you can afford to grant.

WHY NOW · WHY US

# We will make h5i the default place a coding agent runs.

### Why this exists

- PhD research in AI + systems security.
- Dozens of developer conversations on agent autonomy, sandboxing, and what people actually let an agent touch.

### Built already

- 500+ GitHub stars, 20+ releases.
- Five isolation tiers on Linux and macOS, one Rust binary.
- Browser in the box, with human takeover.

### Next 12 weeks

- Prove the browser box in live runs, not fixtures.
- Boot the microVM tier on real virtualization hosts.
- Convert 10 design partners.
- Land first paid pilots.

Give your agent **a whole machine**. **Just not yours.**

h5i.dev

github.com/h5i-dev/h5i